ANSI/IES TM-26-20(R2023)

Illuminating
ENGINEERING SOCIETY

APPROVED METHOD: PROJECTING CATASTROPHIC FAILURE OF LED PACKAGES

AN AMERICAN NATIONAL STANDARD

Approved American National Standard
ANSI

www.ies.org

ANSI/IES TM-26-20(R2023)

APPROVED METHOD: PROJECTING CATASTROPHIC FAILURE OF LED PACKAGES

AN AMERICAN NATIONAL STANDARD

Publication of this report
has been approved by IES.
Suggestions for revisions
should be directed to IES.

Prepared by:
IES Testing Procedures Committee

Approved by the IES Standards Committee, December 19, 2022, as a Transaction of the Illuminating Engineering Society.

Approved April 28, 2023, as an American National Standard.

Published by the Illuminating Engineering Society, 120 Wall Street, New York, New York 10005.

IES Standards are developed through committee consensus and produced by the IES Office in New York. Careful attention is given to style and accuracy. If any errors are noted in this document, they should be forwarded to the Director of Standards, at standards@ies.org or the above address for verification and correction. The IES welcomes and urges feedback and comments.

Printed in the United States of America.

ISBN# 978-0-87995-221-1

DISCLAIMER

IES publications are developed through the consensus standards development process approved by the American National Standards Institute. This process brings together volunteers representing varied viewpoints and interests to achieve consensus on lighting recommendations. While the IES administers the process and establishes policies and procedures to promote fairness in the development of consensus, it makes no guaranty or warranty as to the accuracy or completeness of any information published herein.

The IES disclaims liability for any injury to persons or property or other damages of any nature whatsoever, whether special, indirect, consequential or compensatory, directly or indirectly resulting from the publication, use of, or reliance on this document.

In issuing and making this document available, the IES is not undertaking to render professional or other services for or on behalf of any person or entity. Nor is the IES undertaking to perform any duty owed by any person or entity to someone else. Anyone using this document should rely on his or her own independent judgment or, as appropriate, seek the advice of a competent professional in determining the exercise of reasonable care in any given circumstances.

The IES has no power, nor does it undertake, to police or enforce compliance with the contents of this document. Nor does the IES list, certify, test or inspect products, designs, or installations for compliance with this document. Any certification or statement of compliance with the requirements of this document shall not be attributable to the IES and is solely the responsibility of the certifier or maker of the statement.

AMERICAN NATIONAL STANDARD

Approval of an American National Standard requires verification by ANSI that the requirements for due process, consensus, and other criteria have been met by the standards developer.

Consensus is established when, in the judgment of the ANSI Board of Standards Review, substantial agreement has been reached by directly and materially affected interests. Substantial agreement means much more than a simple majority, but not necessarily unanimity. Consensus requires that all views and objections be considered, and that a concerted effort be made toward their resolution.

The use of American National Standards is completely voluntary; their existence does not in any respect preclude anyone, whether that person has approved the standards or not, from manufacturing, marketing, purchasing, or using products, processes, or procedures not conforming to the standards.

The American National Standards Institute does not develop standards and will in no circumstances give an interpretation to any American National Standard. Moreover, no person shall have the right or authority to issue and interpretation of an American National Standard in the name of the American National Standards Institute. Requests for interpretations should be addressed to the secretariat or sponsor whose name appears on the title page of this standard.

CAUTION NOTICE: This American National Standard may be revised at any time. The procedures of the American National Standards Institute require that action be taken to reaffirm, revise, or withdraw this standard no later than five years from the date of approval. Purchasers of American National Standards may receive current information on all standards by calling or writing the American National Standards Institute.

Prepared by the IES Testing Procedures Committee

KC Fletcher, *Chair*
Andrew Jackson, *Vice Chair*
David N. Randolph, *Secretary*

Members

C. K. Andersen
R. S. Bergman
B. Boudreaux
E. Bretschneider
F. Carpenter
D. Ellis
M. L. Grather
J. N. Hulett
P.-C. Hung
Jianzhong Jiao
M. Kotrebai
B. Kuebler
J. E. Leland
K. M. Liepmann
J. Lockner
S. Longo
L. Loudin
J. P. Marella
S. Mitsuhashi
E. Radkov
M. B. Sapcoe
J. S. Swiernik
J. C. Vollers
Y. Zong

Advisory Members

L. M. Ayers
R. P. Bergin
C. A. Bloomfield
P.-T. Chou
T. E. Cowan
P. Cruz
L. Davis
F. De Caso
J. J. Demirjian
M. E. Duffy
V. Eberhard
P. Elizondo
A. A. Feldman
J. Grandusky
K. J. Hemmi
R. E. HIgley
G. John
J. Juhasz
R. Kelley
J. D. Kramer
K. C. Lerbs
P. McCarthy
G. McKee
C. C. Miller
Y. Ohno
E. Page
D. Park
E. S. Perkins
M. Piscitelli
D. P. Ramer
T. W. Rasinsky
D. Rogers
M. Royer
M. Ruffin
M. Schneider
T. Schneider
A. W. Serres
P. Sharma
R. W. Siddon
D. Spicer
G. A. Steinberg
M. Stevenson
L. Swainston
A. Thorseth

Please refer to the IES Bookstore for possible Errata: www.ies.org.

CONTENTS

1.0 Introduction and Scope

1.1 Introduction

With the completion of *ANSI/IES TM-21-19, Technical Memorandum: Projecting Long Term Maintenance of LED Light Sources*, the LED lighting industry possesses a standard method of obtaining projected long-term luminous flux maintenance information for LED packages.[1] The method is composed of two steps. During the first step, the LED packages shall be tested per *ANSI/IES LM-80-20, Approved Method: Measuring Luminous Flux and Color Maintenance of LED Packages, Arrays and Modules* (and previous versions).[2] The collected measurement data is then used with ANSI/IES TM-21-19 to make luminous flux maintenance projections, including calculations of in-situ temperature interpolations. However, one relevant property is unaddressed by those publications: the catastrophic failure rate for LED packages. Catastrophic failure rate, similar to luminous flux maintenance, is a reliability property that is essential to successfully design LED lamps and luminaires. Combined with luminous flux maintenance, catastrophic LED failure rate information is used by LED lamp and luminaire manufacturers to inform warranty considerations and product application instructions.

In addition to light output decay over time, LED packages experience catastrophic failure, in which no light is produced. These catastrophic failures are typically caused by inadequate product design or process, or improper usage. Catastrophic failure rates for LED packages are much lower than those of other light sources, typically in the range of parts per million hours or parts per billion hours. For practical purposes, LED users, such as LED lamp or luminaire manufacturers, require both luminous flux maintenance life and catastrophic failure rate to adequately assess overall reliability of LED lamps and luminaires.

This document describes three methods for catastrophic-failure rate projections of LED packages.

1.2 Scope

This document describes three methodologies for projecting the catastrophic failure rate of LED packages. This document applies to the LED packages that are defined in *ANSI/IES LS-1-20, Lighting Science: Nomenclature and Definitions for Illuminating Engineering* (see **Section 2.1**). The three methodologies presented are for information only and do not represent a complete set of methodologies in existence; these represent the methodologies that are publicly available, and have been made available, for publication by the IES. The IES does not endorse any of these specific methods, and it is not the intent of this TM to use any of these methods for incorporation into any standards publication.

2.0 Normative References

2.1 ANSI/IES LS-1-20

Illuminating Engineering Society. Lighting Science: Nomenclature and Definitions for Illuminating Engineering. New York: IES; 2020. Online: https://www.ies.org/standards/definitions/. (Accessed 2019 Oct 10).

2.2 NIST/SEMATECH e-Handbook of Statistical Methods, Engineering Statics Handbook

Chapter 8 – Assessing product reliability. Washington, DC: National Institute of Standards and Technology; 2013. Online: http://www.itl.nist.gov/div898/handbook/index.htm. (Accessed 2019 Jul 16).

2.3 SN 29500-12

Siemens Norm. Failure Rates of Components Expected Values for Optical Components. Munich: Siemens; 2008.

2.4 JESD51-51

JEDEC Solid State Technology Association. Implementation of the Electrical Test Method for the Measurement of Real Thermal Resistance and Impedance of Light-Emitting Diodes with Exposed Cooling Surface. Arlington, VA: JEDEC Solid State Technology Association; 2012 Apr.

2.5 JESD85

JEDEC Solid State Technology Association. Methods for Calculating Failure Rates in Units of FITs. Arlington, VA: JEDEC Solid State Technology Association; 2001 (R2014).

2.6 JESD74A

Early Life Failure Rate Calculation Procedure for Semiconductor Components. Arlington, VA: JEDEC Solid State Technology Association; 2007.

3.0 Definitions

3.1 catastrophic LED package failure

The LED package, when energized, does not emit light. Individual die failures in a multi-die package are not considered catastrophic failure so long as the device under test (DUT) continues to produce light.

3.2 failure-in-time (FIT) rate

The frequency with which an LED package catastrophically fails, expressed in failures per one billion (10^9) hours of operation.

4.0 Description of Failure Rates – "Bathtub Curve"

Empirical population failure rates change as LED sources age over time. The general failure rate trends can be summarized by a graph such as the one shown in **Figure 4-1**. Because of the shape of this failure rate curve, it has become widely known as the "bathtub curve."

4.1 Early Failure Period

This is the initial period that begins at time zero when a customer starts to use the product and is characterized by a rapidly decreasing catastrophic failure rate. Typical catastrophic failures that occur during this period may be caused by manufacturing and/or materials defects. The early failure period is also referred to as the *infant mortality period*, from the actuarial origins of the first bathtub curve plots. This decreasing failure rate typically lasts several to a few hundred hours. The impact of the early failure period can often be minimized by screening, combined with a period of product burn-in performed by the LED manufacturer.

4.2 Stable Failure Rate Period

After the early failure period, the failure rate levels off and remains relatively constant in typical cases for the majority of the useful life of the product. This long period of relatively constant failure rate is known as the *stable failure rate period* (also called the *random*, *intrinsic failures period*). It is important to note that most LED packages spend the majority of their lifetime operating in this flat portion of the "bathtub curve."

During the stable failure rate period, the probability of catastrophic failure can be modeled by the exponential cumulative distribution function (CDF) over time:

$$F(t) = 1 - \exp(-\lambda t) \,, \tag{4-1}$$

where:

t = time in hours

λ = failure rate per hour

4.3 Wearout Failure Period

If LED packages remain in use long enough, the catastrophic failure rate begins to increase as materials wear out, and degradation failures occur at an ever-increasing rate. This is the wearout *failure period*.

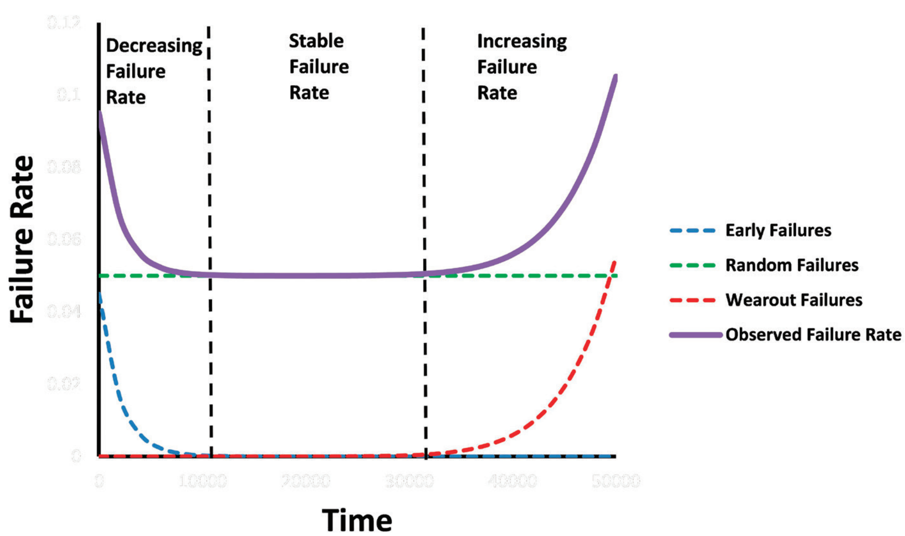

Figure 4-1. Generic "bathtub curve."

5.0 Methods of Failure Rate Projection for the Stable Rate Period

5.1 Prioritization of Catastrophic Failures Analysis and Projection

Whenever possible, the prediction of catastrophic failures over time should be based on experimental observations. In addition, although the Weibull statistical model is used widely, it is important to note that other statistical models, such as a mathematical model based on device physics, may be a better fit to the data.

All "best fit" models shall be based upon statistically significant sample populations, "typical" manufactured parts, well established reliability methodologies, and/or validated mathematical models based upon device physics.

In all cases, projections of cumulative catastrophic failures shall not be extended to other regions of the "bathtub curve" (for example, any projection of random catastrophic failures will not apply to rapid-wearout failures).

Recommendations provided in this section will mainly address the Stable Failure Rate Period—the flat portion of the bathtub curve. It is assumed that this information is most relevant to LED users.

The catastrophic failure rate of LED arrays or modules can be determined either by summation of the catastrophic failure rate of each individual LED package, or directly at the LED array or module level using the methods described in this section.

When actual data are not available and FIT calculations based on wear mechanisms of device physics are used, it should be noted in the report described in **Section 5.4.3**.

5.2 Conversion of All Temperatures to Kelvins

Equation 5-1 shall be used to convert Celsius-scale temperatures to the absolute temperature scale, measured in kelvins (K):

$$T[K] = T[°C] + 273.15 \qquad \textbf{(5-1)}$$

Only values in kelvins shall be used in the calculations shown in the following sections.

5.3 Method 1 – Results of Catastrophic Failure Rate Reported in Table Format

The catastrophic LED failure-in-time (FIT) rate can be reported as numerical results in a table format. For each corresponding value, the LED forward current shall be specified.

Table 5-1 provides an example for using LED junction temperature to report FIT. **Table 5-2** provides an example for using LED case temperature to report FIT.

Table 5-1. Example of FIT Rate* for LED Forward Current vs. Junction Temperature

Forward Current, I_F	Junction Temperature, T_j				
	-10 °C	25 °C	55 oC	85 oC	$T_{j,max}$ °C
$I_{F,\ DC\ maximum}$	100	200	400	800	1,000
$I_{F,\ DC\ nominal}$ 2	40	100	200	400	800
$I_{F,\ DC\ nominal}$ 1	20	40	80	200	600
$I_{F,\ DC\ minimum}$	10	20	40	100	400

* Units failed per one billion hours of operation

Table 5-2. Example of FIT Rate* for LED Forward Current vs. Case Temperature

Forward Current, I_F	Case Temperature, Tc				
	-10 °C	25 °C	55 °C	85 °C	$T_{s,max}$ °C
$I_{F,\ DC\ maximum}$	100	200	400	800	1,000

* Units failed per one billion hours of operation

5.4 Method 2 – Catastrophic Failure Rate Projection Model A

Detailed background information on this method may be found in SN 29500-12 (see Normative **Reference 2.3**).

If the stress factors can be identified, then the LED failure rate (λ) values can be expressed as:

$$\lambda = \lambda_{ref}\,\pi_I\,\pi_T\,, \tag{5-2}$$

where:

λ_{ref} = reference failure rate at reference junction temperature (e.g., 45 °C from empirical data, and 50% of maximum current)

π_I = stress factor for current dependence

π_T = stress factor for temperature dependence

5.4.1 Stress Factor for Temperature Dependence. For a given junction temperature,

$$\pi_T = \exp\left[\frac{E_a}{k_B}\left(\frac{1}{T_{Jr}} - \frac{1}{T_J}\right)\right], \tag{5-3}$$

where:

T_{Jr} = reference junction temperature in kelvins (see Normative **Reference 2.4** for T_{Jr} measurement)

T_J = actual junction temperature in kelvins

k_B = Boltzmann's Constant, 8.6173 x 10^{-5} eV/K

E_a = activation energy for LEDs (e.g., 0.65 eV from empirical data)

In the case where the probability of failure over time can be described by the exponential cumulative distribution function (CDF), then the values of λ_{ref} and λ can be used directly in the exponential CDF.

In the case of the more general Weibull CDF, discussed in **Annex A**, in addition to the value of λ computed from the temperature and current stress factor dependence equations, it is also important to specify the value of β and the value of t for which the λ factor was computed.

5.4.2 Stress Factor for Forward Current Dependence. Given:

$$\frac{I}{I_{max}} \geq 0.5\,,$$

then:

$$\pi_I = \exp\left\{C_4\left[\left(\frac{I}{I_{max}}\right)^{C_5} - \left(\frac{I_{ref}}{I_{max}}\right)^{C_5}\right]\right\}, \tag{5-4}$$

where:

I = operation current in mA (or A),

I_{ref} = reference current in mA (or A),

I_{max} = maximum current in mA (or A),

C_4, C_5 = constants (empirical data). If $I_{ref}/I_{max} = 0.5$, then $C_4 = 1.4$ and $C_5 = 8$.

Otherwise:

$$\pi_I = 1$$

Examples of π_I (values of π_I are between 1 and 4):

$\frac{I}{I_{max}}$	≤ 0.5	0.6	0.7	0.8	0.9	1
π_I	1	1	1.1	1.3	1.8	4

5.4.3 Reporting Results. The following information should be reported:

- Catastrophic failure rate
- Value of λ_{ref}
- Other parameters used in the calculation
- Confidence level

5.5 Method 3 – Catastrophic Failure Rate Projection Model B

Detailed background information may be found in **Normative References 2.5** and **2.6**.

5.5.1 Stress Factor for Temperature Dependence. The FIT formula is applied to project and report the catastrophic failure rate value:

$$\lambda = \frac{\chi^2_\alpha(2r+2)}{2t} \times 10^9\,, \tag{5-5}$$

where:

n = number of devices tested

r = number of catastrophic failures

$\chi^2_\alpha(x)$ = chi-squared distribution for x degrees of freedom and $(1 - \alpha)$ degrees of confidence (confidence level)

t = equivalent device hours, shown as:

$$t = nt_{test}\,\pi_T\,, \tag{5-6}$$

where:

t_{test} = duration of testing

π_T = stress factor for temperature dependence (see **Equation 5-3**)

5.5.2 Stress Factor for Forward-Current Dependence. If the forward-current stress factor, π_I, is considered, an Eyring model may be used:

$$\pi_I = A_T^{\alpha} \exp\left[\frac{E_a}{k_B}\left(B + \frac{C}{T_j}\right) S_1\right], \tag{5-7}$$

where S_1 is a function of forward current, and α, C, and B determine stress between temperature and current stress combinations.

In the case where the probability of catastrophic failure over time can be described by the exponential CDF, then the computed FIT value can be used to estimate the value of λ in the exponential CDF. (See **Annex A**.)

In the case of the more general Weibull CDF, discussed in **Annex A**, in addition to the computed FIT value, it is also important to specify the value of β and the value of t for which the FIT value was computed.

5.5.3 Reporting Results. The following information should be reported:

- Catastrophic failure rate (FIT)
- Model used (data or physical)
- Stress factor for the stated temperature
- Stress factor for the stated forward current (if considered)
- Device-hours (sample size multiplied by testing time)
- Sample size
- Confidence level

ADDITIONAL READING

Liu J, Salmela O, Sarkka J, Morris JE, Tegehall P-E, Andersson C. *Reliability of Microtechnology: Interconnects, Devices and Systems*. New York: Springer-Verlag; 2011.

Annex A – Weibull Analysis for Projection of Catastrophic Failure Rate

Weibull analysis is a common method for projecting catastrophic failure based on experimental data. The 2-parameter Weibull cumulative distribution function (CDF) over time is:

$$F(t) = 1 - \exp\left[-\left(\frac{t}{\eta}\right)^{\beta}\right], \tag{A-1}$$

where:

t = time

η = scale parameter

β = shape parameter

Depending on the value of β, the Weibull distribution can model early failure (when $\beta < 1$), stable failure (with $\beta = 1$), or wearout failure (when $\beta > 1$). The wearout failure type can be further subdivided into early wearout (with $1 < \beta \leq 4$) and rapid wearout (with $\beta > 4$). This analysis therefore can be used for any part of the "bathtub curve" (refer to **Section 4.0**). There is also a Weibull CDF version containing a third (location) parameter, which is not discussed here.

For example, when $\beta = 1$, **Equation A-1** reduces to:

$$F(t) = 1 - \exp\left(-\frac{t}{\eta}\right), \tag{A-2}$$

which yields **Equation 4-1** (see **Section 4.2**) after defining $\lambda = 1/\eta$.

If the value of η is made equal to that of t in **Equation A-2**, this yields:

$$F(\eta) = 1 - \frac{1}{e} \cong 0.632, \tag{A-3}$$

whatever the value of β may be. This shows the physical meaning of η as the time at which 63.2% of the devices are expected to fail. For this reason, η is also called *characteristic life* or $B_{63.2}$ *life*.

It can be seen from **Equation A-2** that:

$$\ln[-\ln(1 - F)] = \beta[\ln t - \ln\eta], \tag{A-4}$$

which defines a straight line with a slope equal to β and an intercept equal to $-\beta\ln\eta$. Here and further below, the time dependence of F is no longer indicated in parentheses, for the sake of simplicity, although it is always implied.

In order to obtain the experimental values of the β and η parameters, at least several catastrophic failures must be observed and the times of their occurrence must be known. A linear fit can then be performed on the function $\ln[-\ln(1-F)]$ vs. $\ln(t)$. Finally, the cumulative failure rate can be projected for future points in time by substituting the β and η values back into the CDF formula for F.

For example, if the cumulative failure rates shown in **Table A-1** were observed at the corresponding times shown , the Weibull plot shown in **Figure A-1** would be obtained, with the cumulative failures projected for future points in time, as shown in **Table A-2**.

The preceding analysis assumes that representative data are available for the cumulative catastrophic failure rates of the LED packages. For catastrophic failure rates

Table A-1. Example of a Weibull Analysis Setup Based on Cumulative Catastrophic Failure Data over Time

F (%)	Time (h)	ln[-ln(1-F)]	ln(t)
0.01	450	-9.21	6.11
0.02	1,118	-8.52	7.02
0.03	1,850	-8.11	7.52
0.04	2,890	-7.82	7.97
0.05	4,335	-7.60	8.37

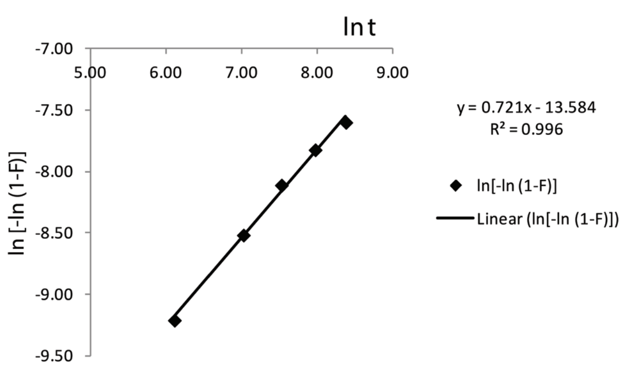

Figure A-1. Example of a Weibull plot for the example data in Table A-1.

Table A-2. Example of Projection of Cumulative Catastrophic Failures over Time*

Time (h)	F_{proj} (%)
5,000	0.06
10,000	0.10
20,000	0.16
30,000	0.21
40,000	0.26
50,000	0.31

* Using the Weibull fitting parameters obtained in Figure A-1.

less than 1%, the function $\ln[-\ln(1\text{-}F)]$ can be simplified to $\ln(F)$, since $\ln(1-x) \cong -x$ when $x \ll 1$.

As a word of caution, extrapolation of Weibull analysis must not be extended beyond the corresponding region of the "bathtub curve" (see **Section 4.0**) within which the data were obtained. For instance, the analysis above clearly will not apply to a wearout situation (which may develop later), since the experimental β value is less than 1.

In order to project other kinds of failures using a Weibull analysis, they must first be observed experimentally, and then corresponding projections can be carried out. A "break," a "corner," or a "dog leg" in the Weibull plot is a visual indication of a change in slope (β), meaning a failure mode change.

Each catastrophic failure mode needs to be processed separately in order to extract the corresponding Weibull parameters, by suspending ("censoring") the samples with other failures. The suspended samples must be accounted for by adjusting the calculated failure percentages accordingly.

Benard's equation can be used as a good estimation of F:

$$F_i = \frac{i - 0.3}{N + 0.4}, \qquad \textbf{(A-5)}$$

where:

F_i is the approximation of F

i is the rank of the failure (1, 2, 3, ...)

N is the number of observations (for example, the total number of samples under test)

In the case of multiple failures recorded at a given time interval, each data point is plotted at the same time with a corresponding rank value.

Data from Weibull analyses may be used to calculate stress factor:

$$AF = \frac{\eta_n}{\eta_a}, \qquad \textbf{(A-6)}$$

where:

η_n = the Weibull scale parameter for the non-accelerated test

η_a = the same parameter for the accelerated one

The β value should not be changed by the used stress and is, therefore, not included in the AF calculation (**Equation A-6**). A significant change in the β value observed as a result of a given stress test would indicate a change in failure mode, rendering the AF approach inapplicable to any such case.

The AF values obtained in this way can be processed further with respect to temperature or current, as shown previously in **Section 5.2** and **Section 5.3**. An example of a temperature related AF calculation from Weibull fits is given in **Table A-3**. The proximity in experimental β values indicates no change in failure mode, so the AF analysis is valid in this case.

Table A-3. Example of Calculation of Temperature-Related $AF(\pi_T)$ Using Weibull Analysis

T_1 (K)	358
T_2 (K)	408
Weibull β at T_1	1.82
Weibull β at T_2	1.88
Weibull η at T_1 (h)	997,220
Weibull η at T_2 (h)	78,542
$AF(\pi_T), T_1 \rightarrow T_2$	12.7
$\frac{E_a}{k_B}$ (K)	7,424

If the β value is well-established for accelerated testing (e.g., based on extensive failure mode analysis), it may be held fixed in the Weibull plots, which would still need to be fitted with respect to η in order to calculate the *AF* values. This allows for relatively fewer data points to be used, as compared to performing two-parameter Weibull fits, which are inherently more sensitive to data noise.

For data sets containing a mix of failed and censored samples, it is possible to generate a maximum likelihood estimate of the characteristic life η, based on a known β value according to **Equation A-7**:

$$\eta = \left[\sum_{i=1}^{N}\left(\frac{t_i^\beta}{r}\right)\right]^{1/\beta}, \quad \textbf{(A-7)}$$

where t_i is the cumulative test time on each unit, r is the number of failed units, and N is the total number of failed and suspended units. This is referred to as *Weibayes analysis*.

Equation A-7 can also provide estimates for situations without observed failures, by setting r to 1 (which amounts to assuming that the first failure is imminent). In this way, a test to first failure can be set up in order to validate a targeted design time (t_d) for any given value of β and for any percentage of cumulative failures. Substituting η from the CDF formula and rearranging for t_i, gives:

$$t_i = \left[\frac{-t_d^\beta}{N\ln(1-F)}\right]^{1/\beta}, \quad \textbf{(A-8)}$$

which gives the required testing time for the Weibayes zero-failure test using N samples. Conversely, the required sample size can be calculated for any fixed test time. As an example, for $\beta = 1$, a design time of 50,000 hours would be validated at the B_{50} ($F = 0.50$) level if there are no catastrophic failures upon operating 12 LEDs for 6,000 hours. Validation at the B_{10} ($F = 0.10$) level under the same remaining conditions would require 79 LEDs.

ADDITIONAL READING

Abernethy RB. The New Weibull Handbook, 5th ed. Utica, NY: Quanterion Solutions Incorporated; 2004.

Dodson B. The Weibull Analysis Handbook, 2nd ed. Milwaukee: ASQ Quality Press; 2006.

INFORMATIVE REFERENCES

1 Illuminating Engineering Society. ANSI/IES TM-21-19, Approved Method: Projecting Long-Term Lumen, Photon, and Radiant Flux Maintenance of LED Light Sources. New York: IES; 2019.

2 Illuminating Engineering Society. ANSI/IES LM-80-20, Approved Method: Measuring Luminous Flux and Color Maintenance of LED Packages, Arrays, and Modules. New York: IES; 2020.References

Lighting Science Standards
Fundamentals, Metrics and Calculations
Lighting Practice Standards
Design, Engineering, and Specifications
Lighting Applications Standards
Design Criteria and Illumination Recommendations
Lighting Measurements and Testing Procedure Standards
Industry Standardization
Roadway and Parking Facility Lighting Standards
Criteria and Illumination Recommendations

Order# ANSI/IES TM-26-20(R2023)
ISBN# 978-0-87995-221-1

www.ies.org